



CHAPTER

1

Atomic Structure

Sr. No.	Concept	Formulas	Other information
1.	Atomic Number (Z)	$Z = \text{Number of protons}$	Mass of a proton = 1.67×10^{-27} kg Charge of a proton = 1.6×10^{-19} C
2.	Mass Number (A)	$A = \text{Number of neutrons} + \text{Number of protons}$	Mass of a Neutron = 1.67×10^{-27} kg Mass of an electron = 9.1×10^{-31} kg Charge of an electron = -1.6×10^{-19} C
3.	Parameters of wave: (a) Wavelength (λ) (b) Frequency (ν) (c) Wavenumber ($\bar{\nu}$)	$\lambda = \frac{c}{\nu}$ $\nu = \frac{c}{\lambda}$ $\bar{\nu} = \frac{1}{\lambda} = \frac{\nu}{c}$	Here, • c is speed of light • $c = 3 \times 10^8$ m/s
4.	Planck's Quantum Theory: (a) Energy of electromagnetic radiation(E)	$E = \frac{hc}{\lambda} = h\nu$ $E = \frac{12400}{\lambda(\text{\AA})} \text{ eV}$ $E = nh\nu = \frac{nhc}{\lambda}$	Here, • h = Planck's constant • $h = 6.626 \times 10^{-34}$ J.s • λ is wavelength of electromagnetic radiation • ν is frequency of electromagnetic radiation • c is speed of light • $c = 3 \times 10^8$ m/s
5.	Photoelectric effect: (a) Energy of photon (E)	$E = W_0 + K.E.$ $h\nu = W_0 + K.E.$ $h\nu = h\nu_0 + K.E.$ $h\nu = h\nu_0 + \frac{1}{2}m_e v^2$	Here, • ' m_e ' is the mass of electron, ' v ' is the velocity of the electron, ν and ν_0 is the frequency and threshold frequency of electromagnetic radiation respectively • h is Planck's constant • W_0 is the work function

Sr. No.	Concept	Formulas	Other information
6.	Bohr's atomic model:		Here,
	(a) Angular momentum	$mvr = \frac{nh}{2\pi}$	• h is Planck's constant • $h = 6.626 \times 10^{-34}$ J·s
	(b) Radius(r)	$r = 0.529 \times \frac{n^2}{Z} \text{ \AA}$	• 'n' is the orbit in which the electron is revolving
	(c) Velocity (v)	$v = 2.188 \times 10^6 \frac{Z}{n} \text{ m/s}$	• 'm' is the mass of electron, • 'v' is the velocity of the electron in the n^{th} orbit,
	(d) Total energy (T.E.)	T.E. = K.E. + P.E. $T.E. = -\frac{KZe^2}{2r}, P.E. = -\frac{KZe^2}{r}, K.E. = \frac{KZe^2}{2r}$ $P.E. = -2K.E., K.E. = -T.E., P.E. = 2T.E.$ $T.E. = -2.18 \times 10^{-18} \frac{Z^2}{n^2} \text{ J/atom}$ $T.E. = -13.6 \times \frac{Z^2}{n^2} \text{ eV/atom}$ $T.E. = -313.6 \times \frac{Z^2}{n^2} \text{ kcal/mol}$ $T.E. = -1312 \times \frac{Z^2}{n^2} \text{ kJ/mol}$	• 'r' is the radius of the n^{th} orbit • 'Z' is atomic number • 'e' is charge on electron • K.E. is kinetic energy of electron • T.E. is total energy of electron • P.E. is potential energy of electron
	(e) Energy difference between n_1 and n_2 energy level	$\Delta E = E_{n_2} - E_{n_1} = 13.6Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ $= I.E. \times \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ $E_n = \frac{-13.6}{n^2} \text{ eV}$	Here, I.E. = Ionization energy of single electron species. • 'n₁' is the initial orbit in which the electron is revolving • 'n₂' is the final orbit in which electron is revolving
	(f) T.E. of any H-like species	$= \text{T.E. of Hydrogen} \times Z^2$ (For same orbit) $= \Delta E$ (For Hydrogen) $\times Z^2$ (For same transition)	• 'Z' is atomic number • 'E_n' is energy of n^{th} energy level
	(g) Energy in n^{th} orbit-for H like species	$= \frac{E_1}{n^2}$ [For same atom]	• 'E_{n₁}' is energy of initial energy level • 'E_{n₂}' is energy of final energy level • T.E. is total energy

Sr. No.	Concept	Formulas	Other information
7.	Hydrogen spectrum: (a) Rydberg's equation (b) Total number of lines in emission spectrum (T.E.L) (c) But for ($n \rightarrow 1$), T.E.L (d) No. of spectral lines in a particular series	$\frac{1}{\lambda} = \bar{\nu} = R_H \left[\frac{1}{n_1^2} - \frac{1}{n_2^2} \right] \times Z^2$ $\bar{\nu} = \frac{1}{\lambda} = R_H Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ $\nu = R_H C Z^2 \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ $E = R_H C Z^2 h \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ $\text{T.E.L} = \frac{(n_2 - n_1)(n_2 - n_1 + 1)}{2}$ $\text{T.E.L} = \frac{n(n-1)}{2}$ $= n_2 - n_1$	Here, $R_H \cong 1.097 \times 10^7 \text{ m}^{-1}$ = Rydberg's constant $\frac{1}{R_H} = 912 \text{ \AA}$ $R_H Ch$ = Energy of 1st orbit of Hydrogen $R_H Ch Z^2$ = Energy of 1st orbit of any Hydrogen like species. n_2 and n_1 are integers ($n_2 > n_1$) 'C' is the speed of the electromagnetic radiation 'n' is the orbit in which the electron is revolving 'n_1' is the initial orbit in which the electron is revolving 'n_2' is the final orbit in which electron is revolving 'Z' is atomic number '$\bar{\nu}$' is wave number
8.	de-Broglie hypothesis	$\lambda = \frac{h}{mv} = \frac{h}{p}$ $\lambda = \frac{h}{\sqrt{2mKE}}$ $\lambda = \frac{h}{\sqrt{2mqV}}$ $\lambda = \sqrt{\frac{150}{V}} \text{ \AA}$	Here, - h is Planck's constant - m is the mass of particle - KE is Kinetic Energy of particle - V is the Potential
9.	Heisenberg uncertainty	$\Delta x \Delta p \geq \frac{h}{4\pi} \text{ or } (\Delta x)(\Delta v) \geq \frac{h}{4\pi m}$	Here, Δx = Uncertainty in position Δv = Uncertainty in velocity Δp = Uncertainty in momentum - m = Mass of particle $\frac{h}{4\pi} = 5.27 \times 10^{-35} \text{ Js}$

Sr. No.	Concept	Formulas	Other information
10.	Quantum number: (a) Principal Quantum number(n) (b) Azimuthal / Secondary/ Subsidiary/ Angular momentum quantum number (l) (c) Orbital angular momentum (L) (d) Spin Quantum Number (m _s or s) (e) Spin angular momentum (S)	Angular momentum = $n \times \frac{h}{2\pi}$ Total number of e ⁻ s in an orbit = $2n^2$ Total number of orbitals in an orbit = n^2 Total number of subshell in an orbit = n Total no. of e ⁻ s in a subshell = $2(2l + 1)$ Total no. of orbitals in a subshell = $(2l + 1)$ $L = \sqrt{l(l+1)} \frac{h}{2\pi}$ Total value of spin in an atom = $\pm \frac{1}{2} \times \text{number of unpaired electrons}$ $S = \sqrt{s(s+1)} \frac{h}{2\pi}$	Here, • h = Planck's constant • $h = 6.626 \times 10^{-34} \text{J}\cdot\text{s}$ • Value of n = 1, 2, 3, 4, 5..... • Value of l = 0, 1.....(n - 1) • Value of s = $\pm \frac{1}{2}$
11.	Schrodinger wave equation	$\frac{\partial^2 \Psi}{\partial x^2} + \frac{\partial^2 \Psi}{\partial y^2} + \frac{\partial^2 \Psi}{\partial z^2} + \frac{8\pi^2 m}{h^2} (E - V) \Psi = 0$	Here, • $\Psi \rightarrow e^-$ wave function • E $\rightarrow$ Total energy • V $\rightarrow$ Potential energy
12.	Radial nodes Angular nodes Total nodes	$n - l - 1$ l $n - 1$	Here, • n is principle quantum number • l is Azimuthal quantum number